FOSTER SCHOOL OF BUSINESS

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AdVantage Digital

Bridging Experimental Rigor with Observational Innovation

Melanie and Xinyu, quantitative marketers at AdVantage Digital, had long relied on randomized controlled trials (RCTs) on Facebook to gauge the causal impact of their digital ad campaigns. Their recent experiments showed that users exposed to ads experienced a meaningful lift in conversions compared to those in the control group. However, as they reviewed the latest results, concerns began to mount over the steep costs involved—not just the direct operational expenses, but also the opportunity cost of withholding ads from control users.

In a brief exchange early in their discussion, Melanie noted, “The RCTs give us the gold standard for measuring causal effects, but every time we assign someone to control, we're missing out on potential conversions. The cost, both in dollars and in lost revenue, is significant.” Xinyu agreed and added, “There’s been so much progress in observational methods recently. Techniques like Inverse Probability Weighting and Double Machine Learning are becoming increasingly popular because they promise to infer causality without the burdens of an experiment. But can they really get close enough to our RCT estimates?”

Over the next couple of weeks, Melanie and Xinyu conceived an intriguing idea. Since their RCT was already yielding a reliable experimental ground truth, they could use the treatment group exclusively as a testing ground for observational methods. By focusing solely on treatment data, they would mimic the real-world scenario where no experiment is run at all. In this environment, the variation in exposure arises naturally—some users are exposed to the ad while others, due to the intricacies of ad delivery, are not. This inherent variation could then be exploited to estimate the causal impact of ad exposure using observational methods.

Professor Amin Sayedi prepared this case to provide material for class discussion. Cases are not intended to serve as endorsements, sources of primary data, or illustrations of effective or ineffective management. Some elements of this case have been disguised or fictionalized for pedagogical reasons and/or to assure confidentiality. No part of this publication may be reproduced, stored in a retrieval system, used in a spreadsheet, or transmitted in any form without the permission of the copyright holder. This case is inspired by Gordon, Brett R., Robert Moakler, and Florian Zettelmeyer. "Close enough? a large-scale exploration of non-experimental approaches to advertising measurement." Marketing Science42.4 (2023): 768-793.

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They reasoned that this approach would allow them to directly compare the estimates from observational methods, such as IPW and Double Machine Learning, against the experimental results. The treatment-only analysis would capture the "natural" scenario where every user is treated, offering a practical framework for evaluating causal inference techniques. With the RCT providing the ground truth, they could assess whether these observational methods could approximate the experimental ATT (Average Treatment effect on the Treated) with acceptable accuracy.

Thus, the stage was set. AdVantage Digital would embark on a pilot study using this treatment-only approach, leveraging the experimental ground truth to rigorously evaluate observational techniques. This narrative, combining advanced statistical methods with practical business considerations, formed the basis for a promising new strategy in digital ad measurement at the firm.

Experimental Setup and Data Description

In our recent Facebook experiment at AdVantage Digital, 2,000,000 users were enrolled to rigorously measure the causal impact of digital ad exposure on conversion rates. The study was designed as a randomized controlled trial, with users randomly assigned to either the treatment group ($Z = 1$) or the control group ($Z = 0$). Approximately 1,000,000 users were assigned to the treatment group, where ads were actively delivered, while the remaining 1,000,000 formed the control group, receiving no ads during the study period.

A unique aspect of this experiment is the application of Ghost ads methodology. Although control users did not actually see any ads, Ghost Ads methodology provided by Meta allowed us to infer whether a control user would have been exposed to an ad if they had been assigned to treatment. In this way, the exposure indicator (W) is available for every user: in the treatment group, W reflects the actual ad exposure, and in the control group, it represents a modeled prediction of exposure. This counterfactual information is critical for comparing outcomes across groups, even though the experimental protocol only directly delivers ads to the treatment group.

The primary outcome of interest, denoted by Y, is a binary indicator that represents whether a user converted—such as making a purchase or signing up for a service—during the experiment. In the treatment group, the probability of conversion depends on whether a user was exposed to the ad ($W = 1$) or not ($W = 0$). For control users, although no ad was delivered, we still capture the modeled exposure indicator, which provides a counterfactual benchmark.

This rich experimental design not only offers a robust causal estimate of ad effectiveness but also creates an opportunity to explore and validate observational causal inference methods. Since the experimental setup delivers a "ground truth" for the Average Treatment effect on the Treated (ATT), we can later assess methods such as Inverse Probability Weighting (IPW) and Double Machine Learning (Double ML) using the treatment group alone. In a real-world scenario without an RCT, all users would be treated; thus, by limiting the observational analysis to the treatment group, we mimic that environment while still leveraging the experimental results as our benchmark.

Detailed Column Descriptions

Variable	Type	Description
age	Integer, 18 to 70	The age of the user, measured in years.
gender	Binary (0 or 1)	A binary indicator of gender. For example, 0 may represent male and 1 female.
activity	Continuous (0–1)	A normalized metric representing the user's engagement on the platform. Higher values indicate more frequent or intense activity.
num_friends	Integer,100-1000	The number of friends or connections a user has on the platform.
mobile_frac	Continuous (0–1)	The proportion of the user's activity that occurs on mobile devices, indicating their preferred mode of access.
prior_conversion	Continuous (0–1)	A normalized indicator of the user's historical conversion behavior. Higher values suggest a greater inherent propensity to convert.
income	Continuous, 25,000 to 200,000	The user's income in U.S. dollars, used as a proxy for socioeconomic status.
urban_index	Continuous (0–1)	A numeric index representing population density or urbanization. Higher values indicate a more urban environment.
Z (Treatment Assignment)	Binary (0 or 1)	The random assignment indicator, where 1 denotes assignment to the treatment group (ad delivery) and 0 indicates assignment to the control group.
W (Exposure Indicator)	Binary (0 or 1)	For treatment group users, W indicates whether the ad was actually delivered. For control group users, although no ad was shown, W reflects our ghost ads prediction—indicating whether a user would have been exposed if assigned to treatment.
Y (Observed Conversion)	Binary (0 or 1)	The observed conversion outcome, with 1 indicating that a user converted (e.g., made a purchase or signed up) during the experiment, and 0 indicating no conversion.

Case Questions

1. **Randomization Check**
   - Choose a few variables, ideally those that you think matter most, and check whether they are the same across treatment and control.
2. **Experimental Lift Calculation: ATT and ITT**
   - **Average Treatment Effect on the Treated (ATT):**
     Estimate the lift for those users who were actually exposed to the ad in the treatment group. Conceptually, ATT represents the causal impact of ad exposure on conversion for users who received the treatment. Calculate

ATT by comparing the observed outcomes of exposed users with an estimate of their counterfactual outcomes (what their conversion rate would have been without the ad) using the experimental data.

- **Intent-to-Treat (ITT):**
  Compute ITT as the simple difference in conversion rates between the treatment and control groups, regardless of whether a user in the treatment group was exposed to the ad. ITT reflects the effect of being assigned to treatment, including any non-compliance, and serves as a useful benchmark for the overall impact of the experiment.

3. **Naive Comparison:**
   Calculate the ATT by directly comparing the conversion rates of exposed (W = 1) versus non-exposed (W = 0) users within the treatment group, without adjusting for any covariates. This naive approach provides an unadjusted measure of lift, which can then be compared to more sophisticated methods.
4. **Adjusted Regression:**
   Estimate the ATT by running a regression that includes the exposure indicator (W) along with relevant covariates as control variables. Report both the absolute lift (the difference in conversion probability attributable to ad exposure) and the relative lift (the absolute lift expressed as a percentage of the baseline conversion rate among non-exposed users).
5. **Inverse Probability Weighting (IPW):**
   Using only treatment data, apply an IPW approach to estimate the ATT by weighting the observations based on the propensity score for exposure. Provide the absolute and relative lift estimates from this method and compare them with the experimental benchmark.
6. **Double Machine Learning (Double ML):**
   Implement Double ML to estimate the ATT using the treatment data. This approach uses machine learning models to estimate nuisance parameters for both the outcome and the exposure, followed by an orthogonalization procedure (via cross-fitting) to obtain a de-biased treatment effect estimate. Report the absolute and relative lift estimates derived from Double ML.
7. **Overall Conclusion and Potential Learnings:**
   Based on your analysis, discuss your overall conclusions regarding the effectiveness of the different methods for estimating ad lift. What did the experimental results reveal, and how did the naive, adjusted regression, IPW, and Double ML estimates compare? Reflect on why the estimates turned out the way they did. Consider factors such as confounding, model specification, and the strength of the relationships between the covariates, exposure, and outcome. What do these findings imply about the potential of observational methods to replicate experimental ground truth in digital advertising contexts?

Appendix A: Challenges in Using Observational Method for Causal Inference in Online Advertising

In online advertising experiment, compliance is perfect in the control group—users assigned to control are never shown campaign ads. In contrast, compliance is one-sided in the treatment group, where exposure to the ad (i.e., receipt of treatment) is not guaranteed. Whether a user in the treatment group is exposed depends on a range of factors related to user behavior, platform-level targeting algorithms, and advertiser bidding strategies. As a result, exposed and unexposed users in the treatment group differ systematically, introducing selection bias into any naive comparison of outcomes. In online advertising settings, this selection bias is particularly pronounced due to three key sources of endogeneity.

User-Induced Endogeneity (Activity Bias)
The first source of bias, often referred to as "activity bias" (Lewis, Rao, and Reiley, 2011), stems from user behavior. To be exposed to an ad, a user must log into Facebook during the campaign window. More active users are both more likely to be exposed and more likely to convert, even in the absence of an ad. For instance, a user on vacation may not visit the platform at all, missing the ad and also being less likely to engage in online purchasing. As a result, unexposed users in the treatment group tend to have lower baseline conversion rates, making them a poor counterfactual for exposed users. This leads to an upward bias in the estimated effect of the ad when comparing exposed and unexposed users without adjustment.

Targeting-Induced Endogeneity
Modern ad delivery systems further amplify selection bias through dynamic targeting. Although targeting criteria for the campaign are fixed at the outset, platforms like Facebook continually optimize ad delivery to meet specific campaign objectives—such as maximizing impressions, click-through rates, or conversions. Over time, the system learns which types of users are most responsive and shifts ad delivery toward those users by upweighting their bids. As a result, users with higher predicted conversion rates are more likely to be exposed. Comparing exposed and unexposed users in this context will tend to overstate ad effectiveness, as the exposed group is selectively made up of higher-propensity converters. In some cases, this bias could run in the opposite direction—for example, if the campaign optimizes for clicks but the advertiser measures success based on purchases. Users who click frequently may not necessarily purchase.

Importantly, this form of bias does not invalidate the experimental design. Facebook's ad delivery system applies the same targeting logic to users in both treatment and control groups. That is, the system still "chooses" which control users would have been shown the ad—except that, at the last moment, the focal ad is swapped out for the next-highest bidder. Because this process is mirrored across treatment and control, the control group remains a valid counterfactual for causal inference.

Competition-Induced Endogeneity

Finally, exposure depends on the advertiser winning the auction for a given impression. This introduces a third layer of endogeneity: competition from other advertisers. Even if the focal advertiser does not introduce selection bias through their own targeting, their exposure pattern may still reflect the bidding strategies of others. For instance, if another advertiser aggressively bids on a high-value segment—say, 18–54-year-old mothers—they may outbid the focal advertiser for that group. As a result, those users will not be exposed to the focal campaign and will instead be overrepresented in the unexposed group. If these users are likely to convert anyway (either through organic behavior or via exposure to complementary ads), the average conversion rate among the unexposed may be higher, potentially leading to downward bias in lift estimates.

In the RCT, we address these various sources of selection bias through random assignment and by tracking exposure status. This allows us to cleanly identify causal effects. In contrast, when using observational methods—such as IPW or Double ML—we discard the control group and attempt to correct for selection bias using only observables within the treatment group. The presence of these multiple sources of endogeneity highlights the importance and difficulty of relying on observational methods to replicate experimental results in online advertising settings.

Appendix B: Lift Estimation Methods

Naive Comparison Approach

A straightforward, or "naive," approach to estimating the causal effect of ad exposure is to simply compare the conversion rates between exposed and non-exposed users within the treatment group. In this method, one calculates the average conversion outcome (Y) for users who were exposed (W = 1) and subtracts the average conversion outcome for those who were not exposed (W = 0). While this method is simple and easy to compute, it does not adjust for any differences in user characteristics that may affect both the likelihood of being exposed and the conversion outcome. Consequently, if exposed and non-exposed users differ systematically in terms of their baseline conversion propensities, the naive comparison will likely overestimate or underestimate the true causal effect of ad exposure. Despite its limitations, the naive approach provides a useful benchmark against which more sophisticated methods, like IPW, can be compared.

Regression Adjustment

Another common approach for estimating the causal effect of ad exposure is to include all relevant user characteristics directly in a regression model. In this method, we use the observed data to compare conversion outcomes between users who were exposed to the ad and those who were not, while simultaneously accounting for differences in factors such as age, gender, engagement, income, and other demographic and behavioral variables. The idea is that by controlling for these characteristics, we can remove the influence of factors that might otherwise skew the results, ensuring that any difference in conversion rates is more likely due to the ad exposure itself.

This regression adjustment approach is appealing because it is both straightforward and intuitive. Rather than relying solely on the raw difference in conversion rates—which can be misleading if exposed and unexposed users differ in important ways—we statistically adjust for these differences within a single model. However, the accuracy of this method depends on having a comprehensive set of covariates and correctly modeling how they influence conversion. If important variables are missing or if the relationships are more complex than assumed, the estimated effect might still be biased.

In practice, this method serves as a useful benchmark against which more advanced techniques can be compared. While it may not capture all the nuances of high-dimensional data like more sophisticated methods do, regression adjustment offers a clear, computationally efficient way to estimate the incremental lift from ad exposure. For further details on the technical aspects of this approach, please refer to the accompanying technical note on "Regression Adjustment for Causal Inference."

Inverse Probability Weighting (IPW)

Inverse Probability Weighting (IPW) is a widely used technique in causal inference that helps correct for selection bias in observational studies. The fundamental idea behind

IPW is to create a pseudo-population where the distribution of covariates is balanced between treated and untreated units. This is achieved by assigning weights to each observation based on the inverse of the probability (propensity score) that the observation received the treatment. In essence, individuals who are less likely to be treated (but actually are) receive higher weights, while those who are more likely to be treated receive lower weights. This reweighting allows for a comparison that more closely approximates a randomized experiment.

In the context of our experiment at AdVantage Digital, we have a rich dataset with multiple covariates such as age, gender, activity, and others. Importantly, the dataset includes an exposure indicator (W), which tells us whether a user was actually exposed to an ad. Although our experiment is an RCT, we are particularly interested in evaluating observational methods by focusing solely on the treatment group. In a real-world scenario without an RCT, all users would be in the treatment group, and we would rely on observational methods to estimate the causal effect of ad exposure.

To apply IPW here, we first estimate the propensity score for each treated user—the probability that they are exposed (W = 1) given their observed covariates. Then, using these propensity scores, we assign weights such that exposed users typically receive a weight of 1, and unexposed users receive a weight of ps/(1 – ps). By performing a weighted regression of the conversion outcome (Y) on the exposure indicator (W) with these weights, we can estimate the Average Treatment effect on the Treated (ATT), or the ad lift. This IPW approach effectively balances the covariate distribution between exposed and unexposed users within the treatment group, allowing us to infer the causal impact of ad exposure and compare it against the experimental ground truth.

Double Machine Learning (Double ML)

Double Machine Learning (Double ML) is an advanced causal inference method that robustly estimates treatment effects by addressing the challenge of high-dimensional nuisance parameters—those unknown functions linking covariates to both treatment assignment and outcomes. In digital marketing, for example, a variable like user activity influences both the likelihood of ad exposure and the probability of conversion. By using flexible machine learning models to capture these relationships, Double ML mitigates bias that could otherwise distort treatment effect estimates.

The approach operates in two stages. First, machine learning models such as LassoCV and LogisticRegressionCV are used to estimate the outcome and treatment models, respectively. This captures the complex interactions among covariates without imposing restrictive assumptions. In the second stage, cross-fitting and orthogonalization techniques are applied to remove the influence of estimation errors in these nuisance parameters. This process effectively isolates the variation in the outcome that is truly attributable to the treatment.

The end result is a doubly robust estimate of the causal impact—such as the Average Treatment Effect on the Treated (ATT)—that is more reliable even when the nuisance models are not perfectly specified. For a more technical discussion and further details on Double ML, please refer to the accompanying note on "Double ML."